Megnath Ramesh

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Education

University of Waterloo, Ph.D. in Electrical and Computer Engineering 2020 – 2025

- **Thesis:** Planning and Replanning Near-optimal Robot Coverage Paths in Partially Unknown Environments.
- **Advisors:** Stephen L. Smith and Baris Fidan
- **Key Courses:** Combinatorial Optimization, Computational Geometry, Deep Learning, Algorithm Design.

University of Alberta, B.Sc. in Electrical Engineering 2013 – 2018

- **GPA:** 3.8/4.0 - Graduated with Distinction.
- **Key Courses:** Multimedia Signal Processing, Continuous/Discrete Time Systems, Control Systems.

Research Appointments

Postdoctoral Fellow (*Vector Faculty Affiliate Researcher*), University of Waterloo 2025 – Present

- **Advisors:** Stephen L. Smith and Baris Fidan
- Developing generative AI (diffusion) models to plan robot paths for long-horizon coverage and exploration tasks in unknown environments.

Ph.D. Researcher, University of Waterloo & Avidbots Corp. 2020 – 2025

- Developed *coverage planning algorithms* for cleaning robots, achieving near-optimal path lengths and minimum turns.
- Tested algorithms using *real-world robots* to demonstrate applicability while encountering environmental uncertainties.
- *Collaborated with the Avidbots engineering team* towards integrating the algorithms with their autonomy stack.
- *Mentored junior researchers* in devising and implementing solutions to difficult robotics problems.

Publications

- **BenchNPIN - Benchmarking Non-Prehensile Interactive Navigation.**
  N. Zhong, S. Caro, A. Iskander, **M. Ramesh**, and S. L. Smith.
  Under review for potential publication at an IEEE conference.
- **Minimum-Length Coverage Path Planning for Grid Environments with Approximation Guarantees.**
  M. Ramesh, F. Imeson, B. Fidan, and S. L. Smith.
  IEEE Robotics and Automation Letters (RA-L), 2025.
- **Approximate Environment Decompositions for Robot Coverage Planning using Submodular Set Cover.**
  M. Ramesh, F. Imeson, B. Fidan, and S. L. Smith.
  IEEE Conference on Decision and Control (CDC), 2024.
- **Anytime Replanning of Robot Coverage Paths for Partially Unknown Environments.**
  M. Ramesh, F. Imeson, B. Fidan, and S. L. Smith.
  IEEE Transactions on Robotics (T-RO), 2024.
- **Optimal Partitioning of Non-Convex Environments for Minimum Turn Coverage Planning.**
  M. Ramesh, F. Imeson, B. Fidan, and S. L. Smith.
  IEEE Robotics and Automation Letters (RA-L), 2022.

Work Experience

Application Software Developer (L1 & L2), ScopeAR 2018 – 2020

- Developed and maintained software for a cross-platform augmented reality (AR) video calling application.
- Led the design for a range of features, including marker-based AR tracking, guest login, and cloud-based AR.

Junior Robotics Developer, VEERUM 2017 – 2018

- Developed a pilot project using autonomous robots for generating digital twins of construction facilities.

Control System Software Developer (Co-op), University of Alberta May – August 2016

- Developed an application to safely control and visually inspect the integrity of a dual thick-wall pipe in a high-pressure pumping system.

Robotics Research Intern (Co-op), École Polytechnique de Montréal — January – April 2016

- Integrated and tested an autonomous mapping algorithm using a Clearpath Husky with a depth camera mounted on a robotic arm.

Teaching Experience

Graduate Teaching Assistant, University of Waterloo — 2021 – 2025

- ECE 406: Algorithm Design and Analysis (Winter 2024 & Winter 2025)
- MTE 484: Digital Control Applications (Fall 2024)
- ECE 484: Digital Control Applications (Fall 2023)
- ECE 380: Analog Control Systems (Spring 2021)

Notable Honors and Awards

- **2023–2025:** 2x Ontario Graduate Scholarship - University of Waterloo ($15k)
- **2021–2025:** 3x UW President's Graduate Scholarship - University of Waterloo ($5k)
- **2021–2022:** Queen Elizabeth II (QEII) Graduate Scholarship - University of Waterloo ($15k)
- **2017:** Jack Neal Prize in Computer and Electrical Engineering - University of Alberta ($500)
- **2015:** Sarah and Martin Gouin Family Scholarship - University of Alberta ($1k)
- **2013–2016:** 3x International Students' Scholarship - University of Alberta ($10k)
- **2013–2016:** 3x Dean's Citation in Engineering - University of Alberta ($5k)
- **2014:** Dean's Research Award - University of Alberta ($500)

Academic Services

Conference reviewer:

- IEEE International Conference on Robotics and Automation (ICRA)
- IEEE/RSJ International Conference on Intelligent Robotics and Systems (IROS)
- IEEE Conference on Control Technology and Applications (CCTA)

Journal reviewer:

- IEEE Transactions on Robotics (T-RO)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Automatic Control (TAC)
- Automatica (Elsevier)

Technical Skills

Languages: Python, C++ (14/17), C, C#, Matlab, Objective-C, Java

Technologies: ROS1/2, ML Libraries (PyTorch, NumPy, scikit-learn), Gymnasium, MuJoCo, Docker, Unity, MQTT, Git

Other Projects

Handi Hand Controller - B.Sc. Capstone Project

Hardware and software for a 3D-printed robotic hand, with computer vision (object recognition) and ROS integration.
In collaboration with BLINC Labs at the University of Alberta

BenchNPIN - Benchmarking Non-Prehensile Interactive Navigation — Project Website

Reinforcement learning (RL) benchmarking tools for robots navigating among movable obstacles.

RoVR the Explorer (*3rd Place Win at HackED 2019*) — GitHub

VR-controlled (teleoperation) ground robot with a point cloud stream to visualize its environment.

Paulie Blart Security Robot (*1st Place Win at SFHacks 2018*) — GitHub

Autonomous security robot to detect intruders (using CNNs). Featured on Global News.